Answers

Worksheets A, B and C

Worksheet A: Algebra and Functions

1. $(3x+2)(x-3)=0$, $x=-\frac{2}{3}$ or $x=3$ [3]

2. $\dfrac{(x-3)(x+3)}{(2x-1)(x+3)}=\dfrac{x-3}{2x-1}$ [3]

3. $2(x-2)+3(x+1)=(x+1)(x-2)\Rightarrow 5x-1=x^2-x-2\Rightarrow x^2-6x-1=0$; $x=3\pm\sqrt{10}$ [5]

4. (a) $\frac{9}{2}$ (b) $y(x-3)=x+4\Rightarrow x(y-1)=3y+4$, $f^{-1}(x)=\dfrac{3x+4}{x-1}$ (c) $fg(x)=\dfrac{2x+3}{2x-4}=3\Rightarrow x=\frac{15}{4}$ [1+3+3]

5. $x^2+(2x-1)^2=10\Rightarrow 5x^2-4x-9=0\Rightarrow(5x-9)(x+1)=0$; $x=\frac{9}{5}, y=\frac{13}{5}$ or $x=-1, y=-3$ [5]

6. $t=\dfrac{v^2-u^2}{2a}$ [3]

7. $(x-7)(x+2)>0$: $x<-2$ or $x>7$ [3]

8. $(4n^2+4n+1)-(4n^2-4n+1)=8n$, a multiple of 8 [3]

9. Second differences 4, so $2n^2$; remainders $3,4,5,6$ give $n+2$; $2n^2+n+2$ [3]

10. $y=\dfrac{k}{x^2}$, $k=36$. (a) $y=1$ (b) $x^2=4$, $x=2$ [2+2]

11. (a) $(x+3)^2-16$ (b) $(-3,-16)$ [2+1]

12. (a) $(2,1)$ (b) $(3,-3)$ (c) $(2,3)$ [1+1+1]

Worksheet B: Geometry, Trigonometry and Vectors

1. (a) $AC^2=8.4^2+11.2^2-2(8.4)(11.2)\cos 64°=113.5$, $AC=10.7$ cm (b) $\frac{1}{2}(8.4)(11.2)\sin 64°=42.3$ cm^2 [4+2]

2. (a) $\sqrt{8^2+6^2+5^2}=\sqrt{125}=11.2$ cm (b) $\tan\theta=\dfrac{5}{10}$, $\theta=26.6°$ [2+2]

3. (a) 62°: angle at the centre is twice the angle at the circumference (b) 28°: $OA=OB$ so triangle OAB is isosceles, $(180-124)\div 2$ [2+2]

4. Length scale factor 1.5. (a) $80\times 1.5^2=180$ cm^2 (b) $240\times 1.5^3=810$ cm^3 [2+2]

5. (a) $\mathbf{b}-\mathbf{a}$ (b) $\overrightarrow{OM}=\mathbf{a}+\frac{1}{2}(\mathbf{b}-\mathbf{a})=\frac{1}{2}(\mathbf{a}+\mathbf{b})$ (c) $\overrightarrow{OP}=\frac{2}{3}\overrightarrow{OM}=\frac{1}{3}(\mathbf{a}+\mathbf{b})$, $\overrightarrow{AP}=\frac{1}{3}(\mathbf{b}-2\mathbf{a})$ [1+2+2]

6. $\cos x=\frac{1}{3}$: $x=70.5°$ or $289.5°$ [3]

7. Sector $\frac{80}{360}\pi(10)^2=69.81$; triangle $\frac{1}{2}(10)(10)\sin 80°=49.24$; segment $=20.6$ cm^2 [4]

8. $\dfrac{(5+\sqrt{2})(3+\sqrt{2})}{9-2}=\dfrac{17+8\sqrt{2}}{7}=\dfrac{17}{7}+\dfrac{8}{7}\sqrt{2}$ [3]

9. Radius gradient $\frac{4}{3}$, tangent gradient $-\frac{3}{4}$: $y-4=-\frac{3}{4}(x-3)$, so $3x+4y=25$ [4]

10. (a) Angle $ABC=245°-140°=105°$; $AC^2=12^2+9^2-2(12)(9)\cos 105°=280.9$, $AC=16.8$ km (b) $\sin A=\dfrac{9\sin 105°}{16.76}$, $A=31.2°$; bearing $=65+31.2\approx 096°$ [4+2]

11. $\dfrac{\sqrt{3}}{2}\times\sqrt{3}=\dfrac{3}{2}$ [2]

© Bridge Up Tuition. Property of Bridge Up Tuition. Free for personal and classroom use; not to be sold or reproduced as your own.

Worksheet C: Number, Probability and Statistics

1. $x = 0.2727\ldots$, $100x = 27.2727\ldots$, $99x = 27$, $x = \frac{27}{99} = \frac{3}{11}$ [2]

2. (a) $84 \div 1.12 = £75$ (b) $4000 \times 1.035^5 = £4750.75$ [2+3]

3. (a) 1.8×10^6 (b) 3×10^{13} [2+2]

4. $a : b = 6 : 10$, $b : c = 10 : 15$, so $6 : 10 : 15$ [3]

5. (a) $3x - 1 = 4$, $x = \frac{5}{3}$ (b) $\dfrac{9y^2}{4x^4}$ [2+3]

6. (a) $1 - \frac{7}{12} \times \frac{6}{11} = 1 - \frac{7}{22} = \frac{15}{22}$ (b) $\frac{4}{n} \times \frac{3}{n-1} = \frac{6}{55} \Rightarrow 660 = 6n(n-1) \Rightarrow n^2 - n - 110 = 0$ (c) $(n-11)(n+10) = 0$, $n = 11$ [3+3+1]

7. (a) FD: 1.2, 1.5, 2.4, 0.7 (b) midpoints $5, 20, 35, 50$: $\dfrac{60 + 600 + 840 + 700}{80} = 27.5$ cm (c) $\frac{5}{10} \times 24 + 14 = 26$ [2+3+2]

8. (a) F only 15, S only 8, both 7, so neither $= 40 - 30 = 10$ (b) $\frac{7}{15}$ [2+2]

9. $12.55 \times 8.45 = 106.0475 \approx 106$ cm^2 [3]

10. (a) $15000 \times 0.86^3 = £9540.84$ (b) Year 6: £6068.51; year 7: £5218.92, so 7 years [2+3]

© Bridge Up Tuition. Property of Bridge Up Tuition. Free for personal and classroom use; not to be sold or reproduced as your own.